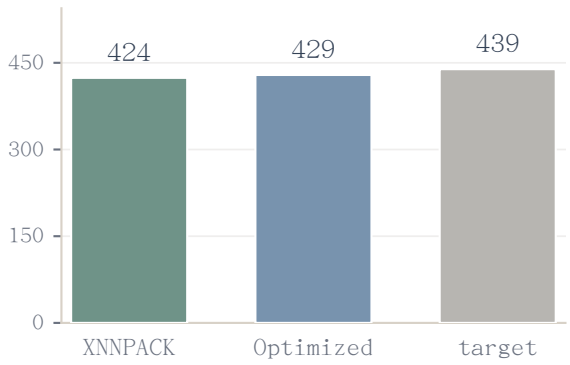
新增 8 类简单 kernel · 主尺寸

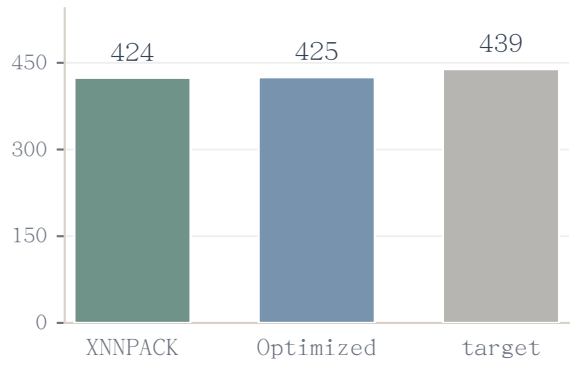
绿色：XNNPACK u8v 手写候选 / 蓝色：公开优化 C / 灰色：target A（修复前）

柱顶为实际目标 cycles，越低越快；各面板纵轴从零起、独立缩放。

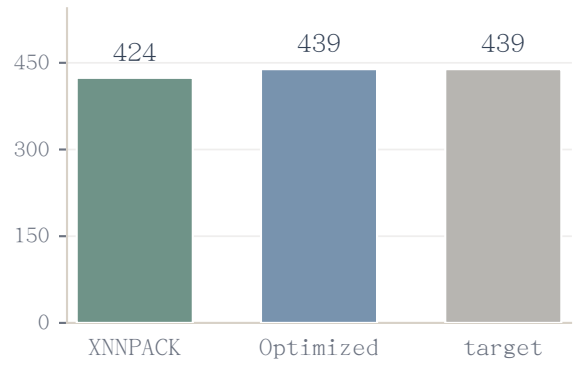
VADD · N=1024



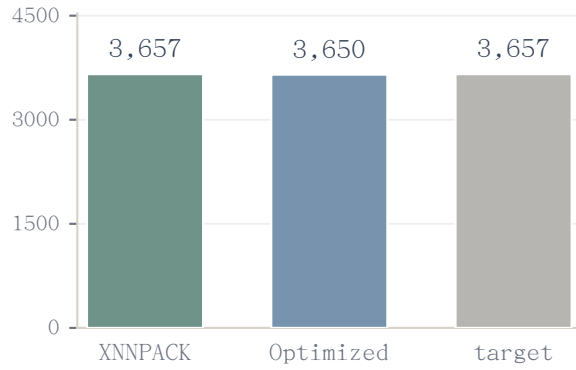
VSUB · N=1024



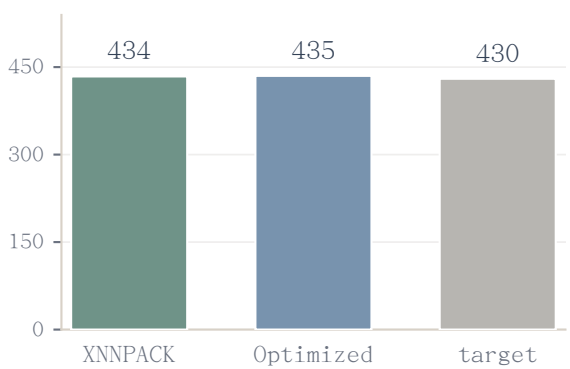
VMUL · N=1024



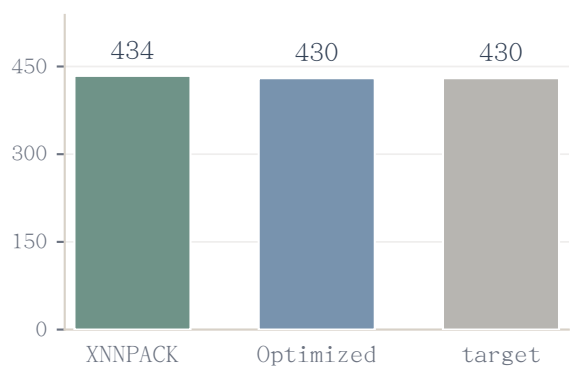
VDIV · N=128



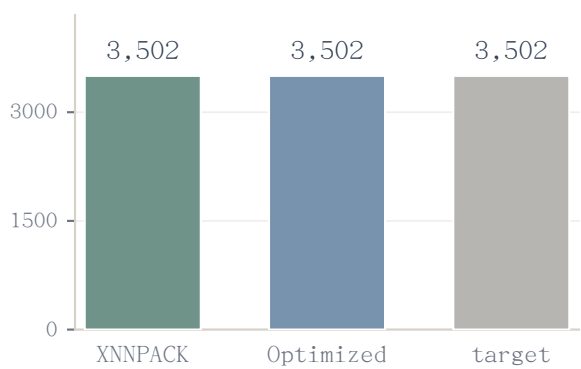
VMIN · N=1024



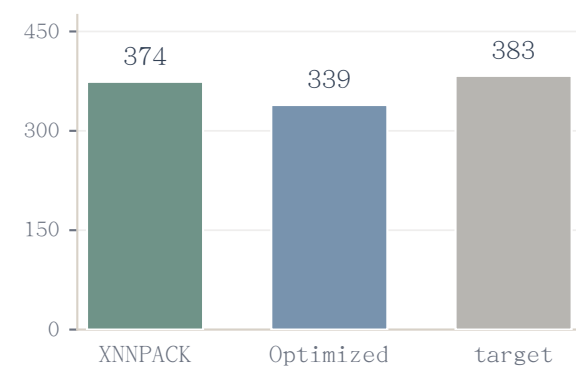
VMAX · N=1024



VSQRT · N=128



Leaky ReLU · N=1024



统一 GCC15.2 / O3 · Shuttle + Saturn RTL · VLEN512 / DLEN256 · 相同输入和调用参数

VDIV/VSQRT 预热 1 次，其余预热 5 次；均计时 3 次取最小；每次逐位检查输出及 canary。原始三次周期另存，没有 cache flush。

仅测试共同有限输入域；除数为正、sqrt 输入非负。小幅差异不作显著性结论；不是新的形式证明。

VMIN/VMAX 冻结修复版 E 另有实测；其余六个 E 与 A 源码一致，不另行重复运行。